

Conclusion

EDA Analysis – Austo Automobile Dataset

- The dataset contains **1,581 customer records** with demographic, financial, and automobile purchase details.
- During data cleaning, missing values and incorrect gender labels (**Fema1, Fem1e**) were corrected successfully.
- Most customers are **Male (around 79%)**, showing that automobile purchases are dominated by male buyers in this dataset.
- The majority of customers are **Married** and belong to the **Salaried profession**, indicating that stable-income families form the main customer base.
- Customers with **higher total salary and partner salary** tend to purchase **higher-priced vehicles**. Income plays a major role in car buying decisions.
- **Sedan** is the most preferred automobile make among customers, followed by other vehicle categories.
- Customers having **house loans or personal loans** are comparatively more cautious while purchasing expensive vehicles.
- Families with working partners generally have a **higher combined income**, which positively affects automobile purchasing power.
- The analysis also shows that **education level and profession influence vehicle preference and spending capacity**.
- Overall, the dataset indicates that automobile purchases are mainly driven by **income level, marital status, profession, and family financial stability**.